United States Patent Application Publication	20250275114
Kind Code	A1
Publication Date	August 28, 2025
Inventor(s)	HSU; Ning-Shuang

CAPACITOR STRUCTURE AND METHOD OF MANUFACTURING THE SAME

Abstract

A manufacturing method of capacitor structures is provided. A template layer is formed over a substrate. A recess is formed in the template layer. A capacitor lower electrode layer is formed on an inner surface of the recess and a top surface of the template layer. The capacitor lower electrode layer has a base located at a bottom of the recess, a side portion extending upwardly from the base to the top surface of the template layer, and a top portion located on the template layer. A ratio of a thickness of the base to a thickness of a topmost side portion of the capacitor lower electrode layer is in a range between about 70% to about 80%. A capacitor dielectric layer is formed in the recess and on the capacitor lower electrode layer. A capacitor upper electrode layer is formed on the capacitor dielectric layer.

Inventors: **HSU; Ning-Shuang (New Taipei City, TW)**

Applicant: **NANYA TECHNOLOGY CORPORATION** (New Taipei city, TW)

Family ID: **1000007713987**

Appl. No.: **18/588713**

Filed: **February 27, 2024**

Publication Classification

Int. Cl.: **H10B12/00** (20230101)

U.S. Cl.:

CPC **H10B12/033** (20230201); **H10B12/31** (20230201);

Background/Summary

BACKGROUND

Field of Invention

[0001] The present disclosure relates generally to a method of manufacturing semiconductor devices and, more specifically, to a method of manufacturing a single-side capacitor structure.

Description of Related Art

[0002] Capacitors continue to have increasing aspect ratios in higher generation integrated circuitry fabrication. For example, dynamic random access memory (DRAM) capacitors now have elevations of from 1 to 3 microns, and widths of less than or equal to about 0.1 micron.

[0003] A common type of capacitor is a so-called container device. A storage electrode of such device is shaped as a container. Dielectric material and another capacitor electrode may be formed within the container and/or along an outer edge of the container, which can form a capacitor having high capacitance and a small footprint.

[0004] Container-shaped storage nodes are becoming increasingly taller and narrower (i.e., are being formed with higher aspect ratios) in an effort to achieve desired levels of capacitance while decreasing the amount of semiconductor real estate consumed by individual capacitors. Unfortunately, high aspect ratio container-shaped storage nodes may decrease the step coverage of the lower electrode of the capacitor.

[0005] Accordingly, it is desired to develop new storage node structures, and new methods for forming storage node structures.

SUMMARY

[0006] It is therefore one objective of the present disclosure to provide a novel method of manufacturing a capacitor structure. This method provides a cost effective and efficient solution to the challenges of achieving high quality and high yield fabrication of DRAM devices with single side capacitor. A capacitor structure obtained by this manufacturing method is also provided in the present disclosure.

[0007] One object of the present disclosure is to provide a method of manufacturing a capacitor structure. The method includes the following steps. A template layer is formed over a substrate. A recess is formed in the template layer. A capacitor lower electrode layer is formed on an inner surface of the recess and a top surface of the template layer. The capacitor lower electrode layer has a base located at a bottom of the recess, a side portion extending upwardly from the base to the top surface of the template layer, and a top portion located on the top surface of the template layer. A ratio of a thickness of the base to a thickness of a topmost side portion of the capacitor lower electrode layer is in a range between about 70% to about 80%. The top portion of the capacitor lower electrode layer is removed to form a capacitor lower electrode. A capacitor dielectric layer is formed in the recess and on the capacitor lower electrode. A capacitor upper electrode layer is formed in the recess and on the capacitor dielectric layer.

[0008] In some embodiments, the capacitor lower electrode layer includes TiSiN.

[0009] In some embodiments, the recess has a depth to width aspect ratio of about 36 to about 46.

[0010] In some embodiments, forming the capacitor lower electrode layer is performed by an atomic layer deposition process.

[0011] In some embodiments, the atomic layer deposition process includes a plurality of cycles, and each cycle includes providing a titanium-containing reactant gas; providing a first purging gas; providing a nitrogen-containing reactant gas; providing a second purging gas; providing a silicon-containing reactant gas; and providing a third purging gas.

[0012] In some embodiments, the titanium-containing is TiCl.sub.4, the nitrogen-containing reactant gas is NH.sub.3, the silicon-containing reactant gas is SiCl.sub.2H.sub.2, and the first purging gas, the second purging gas, and the third purging gas are an inert gas.

[0013] In some embodiments, the inert gas is N.sub.2.

[0014] In some embodiments, the flow rate of the first purging gas is in a range from about 4000

sccm to about 7000 sccm.
[0015] In some embodiments, the flow rate of the second purging gas is in a range from about 4000 sccm to about 7000 sccm.
[0016] In some embodiments, the flow rate of the titanium-containing reactant gas is in a range from about 50 sccm to about 200 sccm.
[0017] In some embodiments, the flow rate of the third purging gas is in a range from about 4000 sccm to about 7000 sccm.
[0018] In some embodiments, the flow rate of the silicon-containing reactant gas is in a range from about 20 sccm to about 60 sccm.
[0019] Another object of the present disclosure is to provide a capacitor structure. The capacitor structure includes a substrate, a template layer, a capacitor lower electrode, a capacitor dielectric layer, and a capacitor upper electrode layer. The template layer is disposed over the isolation layer. The template layer has a recess penetrating the template layer. The capacitor lower electrode is disposed on an inner surface of the recess and surrounded by the template layer. The capacitor lower electrode has a base and a side portion extending upwardly from the base to a top surface of the template layer. A ratio of a thickness of the base to a thickness of a topmost side portion of the capacitor lower electrode is in a range between about 70% and about 80%. The capacitor dielectric layer is disposed in the recess and on the capacitor lower electrode. The capacitor upper electrode layer is disposed in the recess and on the capacitor dielectric layer.
[0020] In some embodiments, a top surface of the capacitor lower electrode is substantially coplanar with the top surface of the template layer.
[0021] In some embodiments, the capacitor structure further includes a conductive layer under the capacitor lower electrode.
[0022] In some embodiments, the base of the capacitor lower electrode is substantially aligned with the conductive layer.
[0023] In some embodiments, the capacitor structure further includes an implant region under the capacitor lower electrode and the conductive layer.
[0024] In some embodiments, the base of the capacitor lower electrode is substantially aligned with the implant region.
[0025] It is to be understood that both the foregoing general description and the following detailed description are by examples, and are intended to provide further explanation of the disclosure as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0026] The disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:
[0027] FIG. **1** is a cross-sectional view schematically depicting a process flow for manufacturing a single side capacitor structure in accordance with one embodiment of present disclosure.
[0028] FIG. **2** is a cross-sectional view schematically depicting a process flow for manufacturing a single side capacitor structure in accordance with one embodiment of present disclosure.
[0029] FIG. **3** is a schematic view schematically depicting a capacitor lower electrode layer formed in a recess by a parameter-adjusted atomic layer deposition process in accordance with one embodiment of present disclosure.
[0030] FIG. **4** is a cross-sectional view schematically depicting a capacitor lower electrode layer formed in a recess without a parameter-adjusted atomic layer deposition process in accordance with one comparative example.
[0031] FIG. **5** is a cross-sectional view schematically depicting a process flow for manufacturing a

single side capacitor structure in accordance with one embodiment of present disclosure.

[0032] FIG. **6** is a cross-sectional view schematically depicting a process flow for manufacturing a single side capacitor structure in accordance with one embodiment of present disclosure.

[0033] FIG. **7** is a cross-sectional view schematically depicting a process flow for manufacturing a single side capacitor structure in accordance with one embodiment of present disclosure.

DETAILED DESCRIPTION

[0034] Reference will now be made in detail to the present embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the description to refer to the same or like parts.

[0035] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0036] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0037] FIG. **1** is a cross-sectional view schematically depicting a process flow for manufacturing a single side capacitor structure in accordance with one embodiment of present disclosure. First, as shown in FIG. **1**, a substrate **110** is provided to serve as a base for forming devices, components, or circuits. In some embodiments, the substrate **110** may be a semiconductor substrate. The substrate **110** may include, consist essentially of, or consist of monocrystalline silicon, and may be referred to as a semiconductor substrate, or as a portion of a semiconductor substrate. The terms “semiconductive substrate,” “semiconductor construction” and “semiconductor substrate” mean any construction including semiconductive material, including, but not limited to, bulk semiconductive materials such as a semiconductive wafer (either alone or in assemblies comprising other materials), and semiconductive material regions (either alone or in assemblies comprising other materials). The term “substrate” refers to any supporting structure, including, but not limited to, the semiconductive substrates described above. Although the substrate **110** in this embodiment is shown to be homogenous, the substrate may include numerous materials in some embodiments. For instance, the substrate **110** may correspond to a semiconductor substrate containing one or more materials associated with integrated circuit fabrication. In such embodiments, such materials may correspond to one or more of refractory metal materials, barrier materials, diffusion materials, insulator materials, etc.

[0038] Referring again to FIG. **1**, an isolation layer **112** is formed over the substrate **110** with source/drain implant regions **114** are shown to be between the isolation layer **112**. The isolation layer **112** may correspond to shallow trench isolation regions in the substrate **110**, and may be filled with any suitable electrically insulated composition or combination of electrically insulated compositions. For instance, in some embodiments the isolation layers **112** may be filled with one or more of silicon dioxide, silicon nitride and silicon oxynitride. The source/drain implant region **114** may include any suitable dopant or combination of dopants and in some embodiments may correspond to an n-type doped region of semiconductor material of substrate **110**. For instance, the substrate **110** may include monocrystalline silicon, and source/drain implant region **114** may correspond to a region of the substrate **110** that is conductively doped with one or both of phosphorus and arsenic. Alternatively, the source/drain implant region **114** may be formed of metal such as tungsten (W) or layers such as TiN/W. The source/drain implant region **114** is one example of an electrical node that may be electrically connected with the base of a storage node. Detailed description will be provided in the following embodiment.

[0039] A template layer **120** is formed over the substrate **110** and covers the isolation layer **112** and the implant region **114**. In some embodiments, the template layer **120** may be a sacrificial layer made of any suitable composition or combination of compositions; and in some embodiments may comprise one or more of silicate glass (for instance, borophosphosilicate glass, phosphosilicate glass, fluorosilicate glass, etc.), spin-on-dielectric, and silicon dioxide formed from tetraethyl orthosilicate (TEOS), or may be a semiconductive layer such as amorphous silicon or polysilicon. Please note that in this embodiment, only one template layer **120** is provided on the substrate **110**. In other embodiment, there may be two or more stacked template layer **120** disposed on the substrate **110**.

[0040] Refer again to FIG. **1**, a plurality of recesses **130** are formed in the template layer **120** for accommodating a capacitor lower electrode layer to be manufactured. The recess **130** extends through the entire thickness of the template layer **120** and exposes the source/drain implant region **114** thereunder. In some embodiments, each recess **130** has a depth to width aspect ratio of about 36 to about 46, such as 37, 38, 39, 40, 41, 42, 43, 44, or 45. In the shown embodiment, the source/drain implant region **114** is configured to be electrically connected with the capacitor lower electrode layer. Thus, a conductive layer **116** is required to be pre-formed over the source/drain implant region **114** to improve electrical coupling between the base of the capacitor lower electrode layer and the conductively-doped implant region **114**. For instance, the conductive layer **116** may be a metal silicide (e.g., titanium silicide, tungsten silicide, etc.) layer formed by silicide process. Alternatively, if the source/drain implant region **114** is formed of metal (ex. W) or metal layers (ex. TiN/W), the conductive layer **116** may be omitted.

[0041] FIG. **2** is a cross-sectional view schematically depicting a process flow for manufacturing a single side capacitor structure in accordance with one embodiment of present disclosure. After providing the template layer **120** and the recesses **130**, a capacitor lower electrode layer **140** is formed an inner surface of the recesses **130** and a top surface **120**S of the template layer **120**. The capacitor lower electrode layer **140** does not fill the recesses **130**, therefore, the smaller recesses **132** are defined in each former recess **130**. In this embodiment, the capacitor lower electrode layer **140** will be used to form the storage node structure with variable thickness. The capacitor lower electrode layer **140** has a base **140***b* located at a bottom of the recesses **130**, a side portion **140***a* extending upwardly from the base **140***b* to the top surface **120**S of the template layer **120**, and a top portion **140**C disposed on the top surface **120**S of the template layer **120**. To be specific, a ratio (namely step coverage) of a thickness **140***bt* of the base **140***b* to a thickness **140***at* of a topmost side portion of the capacitor lower electrode layer **140** is in a range between about 70% to about 80%, such as about 72%, about 74%, about 76%, or about 78%. In some embodiments, the capacitor lower electrode layer **140** includes TiSiN. In one embodiment, the capacitor lower electrode layer **140** consists of TiSiN. In this embodiment, the capacitor lower electrode layer **120** is performed by an atomic layer deposition (ALD) process.

[0042] FIG. **3** is a schematic view schematically depicting a capacitor lower electrode layer formed in a recess by a parameter-adjusted atomic layer deposition process in accordance with one embodiment of present disclosure. In this embodiment, the capacitor lower electrode layer **140** is a TiSiN layer formed using ALD. The reactants may include TiCl.sub.4 **310**, NH.sub.3 **330**, and SiCl.sub.2H.sub.2 (dichlorosilane, DCS) (not shown). The corresponding ALD process includes a plurality of cycles, each cycle including providing a titanium-containing reactant gas, providing a first purging gas, providing a nitrogen-containing reactant gas, providing a second purging gas, providing a silicon-containing reactant gas, providing a third purging gas, and providing a nitrogen-containing reactant gas. In this embodiment, the titanium-containing is TiCl.sub.4, the nitrogen-containing reactant gas is NH.sub.3, the silicon-containing reactant gas is SiCl.sub.2H.sub.2, and the first purging gas, the second purging gas, and the third purging gas are an inert gas. For example, the inert gas may be N.sub.2.

[0043] For example, in each cycle, TiCl.sub.4 may be conducted for about 20 seconds to about 50 seconds, with the flow rate in the range between about 50 sccm and about 200 sccm. According to various embodiments, when the flow rate of TiCl.sub.4 is greater than a certain value such as 200 sccm, it will lead to cost going up. To the contrary, when the flow rate of TiCl.sub.4 is less than a certain value such as 50 sccm, it will cause high aspect ratio capacitor structure having such a thin base thickness during the formation of the capacitor lower electrode layer that the structure will become weak and subject to toppling, twisting and/or breaking from an underlying base. Therefore, the flow rate of TiCl.sub.4 may be such as 60 sccm, 70 sccm, 80 sccm, 90 sccm, 100 sccm, 120 sccm, 140 sccm, 160 sccm, or 180 sccm. NH.sub.3 may be conducted for about 20 seconds to about 50 seconds, with the flow rate in the range between about 4000 sccm and about 7000 sccm. According to various embodiments, when the flow rate of NH.sub.3 is greater than a certain value such as 7000 sccm, it may lead to the pump of the equipment to be unable to withstand too high pressure. To the contrary, when the flow rate of NH.sub.3 is less than a certain value such as 4000 sccm, it will cause high aspect ratio capacitor structure having such a thin base thickness during the formation of the capacitor lower electrode layer that the structure will become weak and subject to toppling, twisting and/or breaking from an underlying base. Therefore, the flow rate of NH.sub.3 may be such as 4500 sccm, 5000 sccm, 5500 sccm, 6000 sccm, or 6500 sccm. N.sub.2 is purging for about 20 seconds to about 50 seconds, with the flow rate in the range between about 4000 sccm and about 7000 sccm. According to various embodiments, when the flow rate of N.sub.2 is greater than a certain value such as 7000 sccm, it will lead to cost going up. To the contrary, when the flow rate of N.sub.2 is less than a certain value such as 4000 sccm, it will cause higher pressure issues in the equipment's chambers. Therefore, the flow rate of N.sub.2 may be such as 4500 sccm, 5000 sccm, 5500 sccm, 6000 sccm, or 6500 sccm. The operating conditions of SiCl.sub.4H.sub.2 remain the same as the original.

[0044] The reaction chamber may be maintained a pressure of about 3 Torr to about 6 Torr, such as about 3.2 Torr, about 3.4 Torr, about 3.6 Torr, about 3.8 Torr, about 4.0 Torr, about 4.2 Torr, about 4.4 Torr, about 4.6 Torr, or about 4.8 Torr. The reaction chamber may also be maintained at an elevated temperature of about 500° C. to about 600° C., such as about 510° C., about 520° C., about 530° C., about 540° C., about 550° C., about 560° C., about 570° C., about 580° C., or about 195° C.

[0045] It should be noted that the flow rate of TiCl.sub.4 and/or NH.sub.3 may be adjusted by increasing 2 times of the original flow rate of TiCl.sub.4 and/or NH.sub.3. When the flow rate of TiCl.sub.4 and/or NH.sub.3 are increased in the deposition chamber, the reactant gases (such as TiCl.sub.4 and NH.sub.3) may allow reaching the bottom of the recess **130** more easily, thereby increasing the thickness of the deposition thin film. In addition, the flow rate of the purge gas **320** may be adjusted by increasing 1.5 times of the original flow rate of the purge gas **320** (such as N.sub.2). When the flow rate of N.sub.2 is increased in the deposition chamber, the purge gas **320** may help remove excess reactant gas and by-products, thereby improving the quality of the deposition thin film. The deposited capacitor lower electrode layer **140** (TiSiN layer) is shown as FIG. **2**. The thickness **140***at* of the topmost side portion **140***a* of the capacitor lower electrode layer **140** is in the range between about 14.40 μm to about 14.55 μm, such as 14.41 μm, 14.43 un, 14.45 μm, 14.47 μm, 14.49 μm, 15.51 μm, or 15.53 μm. The thickness **140***bt* of the base **140***b* of the capacitor lower electrode layer **140** is in the range between about 10.75 μm to about 11.15 μm, such as 10.77 μm, 10.79 un, 10.80 μm, 10.85 μm, 10.90 μm, 10.95 μm, 11.00 μm, 11.05 μm, or 11.10 μm. It may be understood that a ratio (namely step coverage) of the thickness of the base to the thickness of the topmost side portion is in a range between about 70% and about 80%, such as 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, or 79%.

[0046] FIG. **4** is a cross-sectional view schematically depicting a capacitor lower electrode layer formed in a recess without a parameter-adjusted atomic layer deposition process in accordance with one comparative example. In this comparative example, in each cycle, TiCl.sub.4 may be conducted for about 20 seconds to about 50 seconds, with the flow rate in the range between about 2000 sccm and about 4000 sccm. NH.sub.3 may be conducted for about 20 seconds to about 50 seconds, with the flow rate in the range between about 2000 sccm and about 4000 sccm. N.sub.2 is purging for about 20 seconds to about 50 seconds, with the flow rate in the range between about 2000 sccm and about 4000 sccm. SiCl.sub.2H.sub.2 may be conducted for about 20 seconds to about 50 seconds, with the flow rate in the range between about 20 sccm and about 60 sccm. The deposited capacitor lower electrode layer **141** (TiSiN layer) is shown as FIG. **4**. The thickness of the upper sidewall **141***a* of the capacitor lower electrode layer **141** is in the range between about 5 nm to about 15 nm. The thickness of the base **141***b* of the capacitor lower electrode layer **141** is in the range between about 5 nm to about 15 nm. The step coverage shown in FIG. **4** is in a range between about 53% and about 54%. Because the recess **130** has a high depth to width aspect ratio, it is difficult for the reactant gas to reach the bottom of the recess for deposition, so that the profile of the capacitor lower electrode layer **140** appears as shown in FIG. **4**. This situation may also lead to high aspect ratio capacitor structure having such a thin base thickness during the formation of the capacitor lower electrode layer that the structure will become weak and subject to toppling, twisting and/or breaking from an underlying base.

[0047] Comparing with the comparative example, increasing the flow rate of TiCl.sub.4, NH.sub.3, and/or N.sub.2 in the present disclosure may improve the step coverage while maintain a high productivity.

[0048] FIG. **5** is a cross-sectional view schematically depicting a process flow for manufacturing a single side capacitor structure in accordance with one embodiment of present disclosure. After forming the capacitor lower electrode layer **140**, a chemical mechanical polishing (CMP) process is performed to remove the top portion **140**C of the capacitor lower electrode layer **140** on the template layer **120** to form a capacitor lower electrode **142** embedded in the template layer **120**. The capacitor lower electrode **142** includes a base **142***b* along the bottom which is electrically connected with the source/drain implant region **114** (through the conductive layer **116**) thereunder, and two side portions **142***a* extending upwardly from the base **142***b*. The top surface of the capacitor lower electrode **142** is coplanar with the top surface of the template layer **120**. Although there appear to be two separate side portions **142***a* along the cross-section of the view of FIG. **5**, such side portions **142***a* may be a part of a single sidewall structure when considered in three dimensions, such as a single circular cylinder when viewed from above.

[0049] The side portions **142***a* in the cross-sectional view of FIG. **5** have a substantially constant thickness from the base of the sidewalls to the tops of the sidewalls. The term “substantially constant thickness” means that the thickness is uniform to within tolerances imposed by the fabrication process utilized to form the capacitor lower electrode **142**. In some embodiments, the thickness of the side portion **142***a* of the capacitor lower electrode **142** may be within a range of from about 40 Å to about 100 Å, and may be, for example, about 50 Å, 60 Å, 70 Å, 80 Å, or 90 Å. In some embodiments, the thickness may vary from the top of the sidewalls to the base of the sidewalls, with the upper portion of the sidewall being thicker than the lower portion of the sidewall due to difficulties associated with the uniform deposition of the capacitor lower electrode layer **140** within a high aspect ratio opening during formation of the capacitor lower electrode **142**.

[0050] In the shown embodiment, the source/drain implant region **114** is electrically connected to the capacitor lower electrode **142**. In some embodiments, the capacitor lower electrode **142** is ultimately incorporated into a capacitor, and such capacitor is ultimately connected to a transistor to form a DRAM unit cell. Thus, the source/drain implant region **114** may connect to a transistor gate that gatedly couples the source/drain implant region **114** to another source/drain implant region (not shown). The transistor gate may be part of an access line (i.e., a word line), and the other source/drain region may be connected to a bit line. Accordingly, the capacitor lower electrode **142** may be uniquely addressed through the combination of the bit line and the access line. The shown capacitor lower electrode **142** may be one of a large plurality of storage node structures that are subjected to identical processing during fabrication of a DRAM array.

[0051] FIG. **6** is a cross-sectional view schematically depicting a process flow for manufacturing a single side capacitor structure in accordance with one embodiment of present disclosure. As shown in FIG. **6**, after the capacitor lower electrode **142** is planarized, a capacitor dielectric layer **150** is formed in the recess **132** and on the capacitor lower electrode **142**. The capacitor dielectric layer **150** is formed conformally along the inner surfaces of the capacitor lower electrode **142** including the side portions **142***a* and the base **142***b* and the top surface of the template layer **120**. In some embodiments, the capacitor dielectric layer **150** may include any suitable composition or combination of compositions, such as one or both of silicon nitride and silicon dioxide. The capacitor dielectric layer **150** may be formed utilizing any suitable methods, including, for example, one or more of atomic layer deposition (ALD), chemical vapor deposition (CVD), and physical vapor deposition (PVD).

[0052] FIG. **7** is a cross-sectional view schematically depicting a process flow for manufacturing a single side capacitor structure in accordance with one embodiment of present disclosure. A capacitor upper electrode layer **160** is then formed in the recess and on the capacitor dielectric layer **150**. The capacitor upper electrode layer **160** fills up the recess formed in the side portions **142***a* and the base **142***b* of the capacitor lower electrode **142** and covers the entire surface of the template layer **120**. In some embodiments, the capacitor upper electrode layer **160** may include any suitable composition or combination of compositions, such as one or more of various metals (for instance, titanium, tungsten, etc.), metal-containing compositions (for instance, metal nitride, metal silicide, etc.) and conductively-doped semiconductor materials (for instance, conductively-doped silicon, conductively-doped germanium, etc.). The capacitor upper electrode layer **160** may be formed utilizing any suitable methods, including, for example, one or more of atomic layer deposition (ALD), chemical vapor deposition (CVD), and physical vapor deposition (PVD).

[0053] The capacitor structure **70** shown in FIG. **7** includes a substrate **110**, a template layer **120**, a capacitor lower electrode **142**, a capacitor dielectric layer **150**, and a capacitor upper electrode layer **160**. The template layer **120** is disposed over the substrate **110**, in which the template layer **120** has a recess **130** penetrating the template layer **120**. In some embodiments, the capacitor structure **70** further includes an isolation layer **112** disposed between the substrate **110** and the template layer **120**. The various features of each element (the substrate **110**, the isolation layer **112**, and the template layer **120**) as shown in FIG. **7** have been described hereinbefore, and the details are not repeated herein.

[0054] The capacitor lower electrode **142** is disposed on an inner surface of the recess **130** and surrounded by the template layer **120**. The capacitor lower electrode **142** has a base **142***b* and a side portion **142***a* extending upwardly from the base **142***b* to a top surface **120**S of the template layer **120**. A ratio (namely step coverage) of a thickness **142***bt* of the base **142***b* to a thickness **142***at* of a topmost side portion **142***a* of the capacitor lower electrode **142** is in a range between about 70% and about 80%, such as 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, or 79%. In some embodiments, a top surface **142**TS of the capacitor lower electrode **142** is coplanar with the top surface **120**S of the template layer **120**. The capacitor dielectric layer **150** is disposed in the recess **130** and on the capacitor lower electrode **142**. The capacitor upper electrode layer **160** is disposed in the recess **130** and on the capacitor dielectric layer **150**.

[0055] The capacitor structure **70** may further include a conductive layer **116** under the capacitor lower electrode **142**. In some embodiments, the base **142***b* of the capacitor lower electrode **142** is substantially aligned with the conductive layer **116**. The term “substantially aligned” used herein refers to a vertical projection of the conductive layer **116** overlapped with a vertical projection of the base **142***b* of the capacitor lower electrode **142**. In addition, the capacitor structure **70** may further include an implant region **114** under the capacitor lower electrode **142** and the conductive layer **116**. In some embodiments, the base **142***b* of the capacitor lower electrode **142** is substantially aligned with the implant region **114**.

[0056] The embodiment of FIG. **7** shows the capacitor lower electrode **142** incorporated into a capacitor including such storage node in combination with the capacitor dielectric layer **150** and the capacitor upper electrode layer **160**. This single side capacitor may be utilized in combination with a transistor (which may correspond to the circuit in the substrate **110**) to form a DRAM unit cell, and such unit cell may be representative of a large number of unit cells simultaneously formed and incorporated into a DRAM array.

[0057] By increasing the reactant gas flow rate of TiCl.sub.4 and/or NH.sub.3, the gas pressure in the deposition chamber is enhanced, thereby allowing the reactant gas reaching the bottom of the recess having a high aspect ratio. In addition, increasing the purge gas flow rate of N.sub.2 helps remove excess reactant gas and by-products, thereby improving the quality of the deposition thin film. The method of manufacturing the capacitor structure of the present disclosure provides a cost-effective and efficient solution to the changes of achieving high quality and high yield fabrication of DRAM devices with single side capacitor structures.

[0058] Although the present disclosure has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

[0059] It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present disclosure without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the present disclosure cover modifications and variations of this disclosure provided they fall within the scope of the following claims.

Claims

1. A method of manufacturing a capacitor structure, comprising: forming a template layer over a substrate; forming a recess in the template layer; forming a capacitor lower electrode layer on an inner surface of the recess and a top surface of the template layer, wherein the capacitor lower electrode layer has a base located at a bottom of the recess, a side portion extending upwardly from the base to the top surface of the template layer, and a top portion located on the top surface of the template layer, wherein a ratio of a thickness of the base to a thickness of a topmost side portion of the capacitor lower electrode layer is in a range between about 70% to about 80%; removing the top portion of the capacitor lower electrode layer to form a capacitor lower electrode; forming a capacitor dielectric layer in the recess and on the capacitor lower electrode; and forming a capacitor upper electrode layer in the recess and on the capacitor dielectric layer.

2. The method of claim 1, wherein the recess has a depth to width aspect ratio of about 36 to about 46.

3. The method of claim 1, wherein the capacitor lower electrode layer comprises TiSiN.

4. The method of claim 1, wherein forming the capacitor lower electrode layer is performed by an atomic layer deposition process.

5. The method of claim 4, wherein the atomic layer deposition process comprises a plurality of cycles, and each cycle comprises: providing a titanium-containing reactant gas; providing a first purging gas; providing a nitrogen-containing reactant gas; providing a second purging gas; providing a silicon-containing reactant gas; and providing a third purging gas.

6. The method of claim 5, wherein the titanium-containing is TiCl.sub.4, the nitrogen-containing reactant gas is NH_3, the silicon-containing reactant gas is $SiCl_2H_2$, and the first purging gas, the second purging gas, and the third purging gas are an inert gas.

7. The method of claim 6, wherein the inert gas is N_2.

8. The method of claim 5, wherein a flow rate of the first purging gas is in a range from about 4000 sccm to about 7000 sccm.

9. The method of claim 5, wherein a flow rate of the second purging gas is in a range from about 4000 sccm to about 7000 sccm.

10. The method of claim 5, wherein a flow rate of the titanium-containing reactant gas is in a range from about 50 sccm to about 200 sccm.

11. The method of claim 5, wherein a flow rate of the third purging gas is in a range from about 4000 sccm to about 7000 sccm.

12. The method of claim 5, wherein a flow rate of the silicon-containing reactant gas is in a range from about 20 sccm to about 60 sccm.

13. A capacitor structure, comprising: a substrate; a template layer disposed over the substrate, wherein the template layer has a recess penetrating the template layer; a capacitor lower electrode disposed on an inner surface of the recess and surrounded by the template layer, the capacitor lower electrode has a base and a side portion extending upwardly from the base to a top surface of the template layer, wherein a ratio of a thickness of the base to a thickness of a topmost side portion of the capacitor lower electrode is in a range between about 70% and about 80%; a capacitor dielectric layer disposed in the recess and on the capacitor lower electrode; and a capacitor upper electrode layer disposed in the recess and on the capacitor dielectric layer.

14. The capacitor structure of claim 13, wherein a top surface of the capacitor lower electrode is substantially coplanar with the top surface of the template layer.

15. The capacitor structure of claim 13, further comprising a conductive layer under the capacitor lower electrode.

16. The capacitor structure of claim 15, wherein the base of the capacitor lower electrode is substantially aligned with the conductive layer.

17. The capacitor structure of claim 15, further comprising an implant region under the capacitor lower electrode and the conductive layer.

18. The capacitor structure of claim 17, wherein the base of the capacitor lower electrode is substantially aligned with the implant region.
